

# COVARIATE SHIFT COMPOUNDING ERROR FLYWHEEL & RECOVERY DYNAMICS

Quadratic Divergence  $O(T^2\epsilon)$  in Naive Behavioral Cloning vs Linear Bounded Tube  $O(T\epsilon)$  in DAgger Corrective Aggregation

## × NAIVE BEHAVIORAL CLONING (OPEN-LOOP DRIFT FLYWHEEL)

Error:  $O(T^2\epsilon)$

### 1. Single-Step Perturbation $\epsilon$

Friction / torque / sensor noise

### 2. State Exits Support

$s_{t+1} \notin \text{supp}(d^{\pi^*})$

### 4. Amplified Deviation

Pushed further off-manifold

### 3. Arbitrary Action $\hat{a}$

Zero constraint from ERM loss

### Mathematical Compounding Mechanism:

$$E[\text{Error}_{BC}] \leq \sum t \cdot \epsilon = [T(T+1)/2] \cdot \epsilon = O(T^2\epsilon)$$

Example ( $T = 500$  steps,  $\epsilon = 0.01$ )  $\Rightarrow$  Cumulative Error Factor = 2,500×

## ✓ CORRECTIVE AGGREGATION / DAgger (STABILIZING FLYWHEEL)

Error:  $O(T\epsilon)$

### 1. Policy Visits $s \sim d_{\text{policy}}$

Learner visits perturbed state

### 2. Expert Labels $\pi^*(s)$

Recovery action demonstrated

### 4. Closed-Loop Recovery

Restoring torque restores state

### 3. Aggregate to Dataset $D$

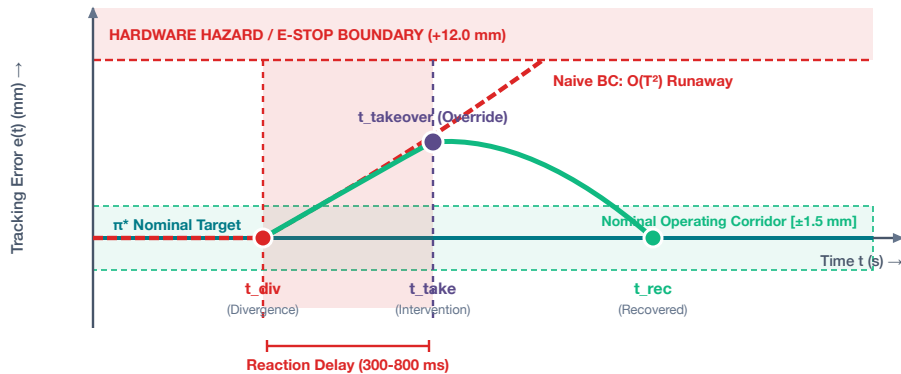
$\text{supp}(D) \supseteq \text{recovery corridor}$

### Mathematical Linear Bounded Corridor:

$$E[\text{Error}_{CA}] \leq T \cdot \epsilon = O(T\epsilon) \text{ [Ross \& Bagnell 2011]}$$

Example ( $T = 500$  steps,  $\epsilon = 0.01$ )  $\Rightarrow$  Cumulative Error Factor = 5.0× (500× reduction!)

## TRAJECTORY DIVERGENCE & INTERVENTION CUT BOUNDARY



### THE 3 DATA CURATION LAWS FOR PHYSICAL INTERVENTIONS:

- 1. Truncate Autonomous Rollout at  $t_{div}$  (NOT  $t_{takeover}$ ):**  
Cut rollout where policy drifted. Prevents neural networks from learning that approaching a hazard precedes goal completion.
- 2. Discard Corrupted Drift Interval [ $t_{div}$ ,  $t_{takeover}$ ]:**  
Purge 30-80 frames logged during human reaction delay. These transitions contain divergent commands falsely logged as nominal.
- 3. Ingest Recovery Demonstration [ $t_{takeover}$ ,  $t_{rec}$ ] into DAgger Buffer:**  
Treat rescue as expert feedback on policy distribution  $d_{\text{policy}}$ , expanding support to recovery corridor and bounding error to  $O(T\epsilon)$ .